

- 14** **D** The time required for the object to move from P to Q, half a complete oscillation, is half a period, $\frac{T}{2}$. Thus, **A** and **B** are eliminated since the options are more than $\frac{T}{2}$.

4

Consider the time taken t required for the object in Simple Harmonic Motion (SHM) with amplitude A to move from P to X, thus

$$\frac{A}{2} = A \cos\left(\frac{2\pi}{T}t\right) \rightarrow t = \frac{T}{6}$$

and the required answer is given by

$$\frac{T}{2} - \frac{T}{6} = \frac{T}{3}$$